9 Francesco carried out a survey about the ages of the people in his office.

The table shows information about his results.

Age (a years)	Cumulative frequency
$20 < a \leqslant 30$	10
$20 < a \leqslant 40$	26
$20 < a \leqslant 50$	58
$20 < a \leqslant 60$	66
$20 < a \leqslant 70$	70

(a) On the grid opposite, draw a cumulative frequency graph for this information.

(2)

Plot upper class boundaries against cumulative totals:

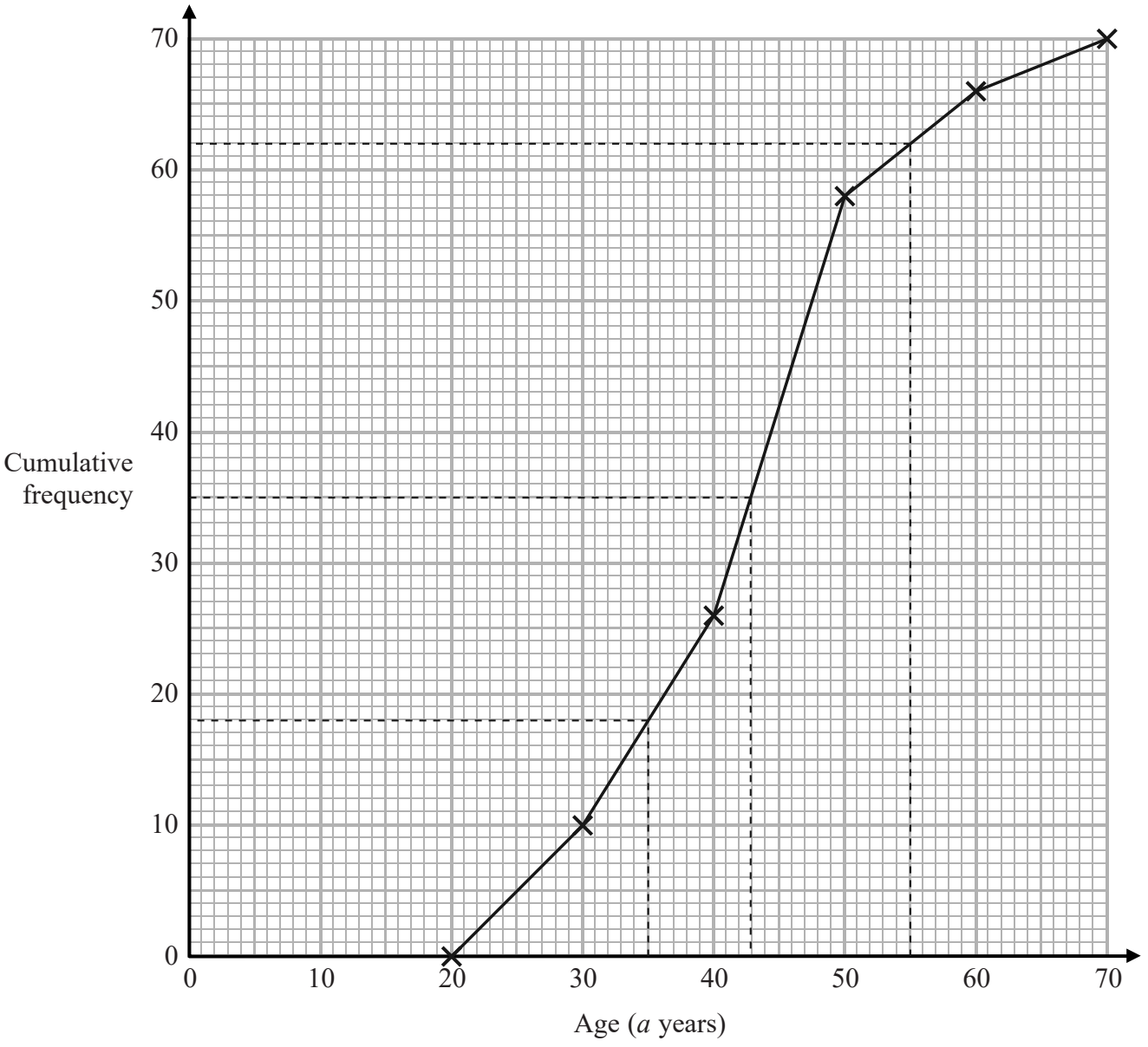
1 mark
Correct boundary points

(20,0), (30,10), (40,26), (50,58), (60,66), (70,70).

Join in order: cumulative frequency cannot decrease.

1 mark
Join without decreasing

Use 70÷2=35 for the median on the opposite page.



(b) Use your graph to find an estimate for the median age.

70÷2=35th → about43 years.

1 mark
Read median at35

.....years

(1)

Francesco says,

"More than 60% of the people in the office are between 35 and 55 years old."

(c) Use your graph to determine if Francesco is correct.

At35:18; at55:62, so62-18=44 people.

44÷70 ×100 = 62.9% >60%.

Yes, Francesco is correct.

1 mark
Read at35 and55

1 mark
44 out of70

1 mark
62.9% exceeds60%

(3)

(Total for Question 9 is 6 marks)